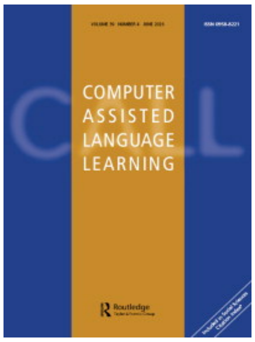

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Comparative analysis of high and low performers' use of generative AI in EFL academic writing: behavioral patterns and perceptions

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ABSTRACT

Despite the growing integration of GenAI tools into academic writing, little is known about how learners with different performance levels engage with them. This study examines how EFL learners use GenAI during an academic writing task, conceptualizing AI affordances and constraints through a self-regulated learning (SRL) framework that focuses on learners' questioning and feedback adoption behaviors. Thirty-seven learners initially participated; two who did not use GenAI were excluded, resulting in a final sample of 35. Based on writing scores, the top 40% ($n = 14$, Group H) and bottom 40% ($n = 14$, Group L) were selected for comparison. Data sources included assignments, screen recordings, GenAI chat logs, and post-task interviews. Descriptive and inferential statistics were used to compare questioning and adoption patterns, complemented by qualitative content analysis. Both groups used GenAI with similar frequency but differed in how they engaged with it. Group H primarily asked language-focused questions related to translation refinement and academic expression, whereas Group L relied more on GenAI for content generation and comprehension support. Group H also revised AI outputs more extensively. Although both groups viewed language editing as GenAI's primary strength, they differed in how they interpreted its limitations: Group H attributed problems to prompting strategies, whereas Group L emphasized technical constraints. Based on these patterns, the study proposes pedagogical principles for AI-mediated EFL writing.

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EFL academic writing; generative AI; self-regulated learning; questioning behavior; adoption behavior

1. Introduction

Academic writing plays a central role in scholarly communication and academic advancement (Garvey, 1979). Producing effective academic texts requires writers to coordinate linguistic accuracy, disciplinary knowledge, and rhetorical organization through iterative planning, drafting, and revision (Allen & McNamara, 2017). For learners of English as a Foreign Language (EFL), these demands are particularly challenging. Even when learners meet formal language proficiency requirements,

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limited control over academic registers and disciplinary conventions often constrains their ability to articulate complex ideas in writing (Manalo & Sheppard, 2016; Rakedzon & Baram-Tsabari, 2017).

Recently, generative artificial intelligence (GenAI) has reshaped writing support by offering real-time, interactive assistance across multiple stages of the writing process, addressing both higher-level concerns (e.g. argumentation and organization) and lower-level linguistic issues (Escalante et al., 2023; Wang, 2025). As these tools become increasingly accessible and normalized in academic contexts, research attention has shifted from whether GenAI should be used to how learners actually engage with it during authentic writing tasks. Emerging evidence suggests that even under comparable task conditions, learners differ markedly in how they interact with GenAI and in the quality of their resulting texts (Johnson, 2024), raising questions about the sources of such variation.

Self-regulated learning (SRL) offers a well-established theoretical lens for examining complex learning tasks, including academic writing (Zimmerman, 2002). From an SRL perspective, effective writing depends not only on linguistic knowledge, but also on learners' ability to plan, monitor, and evaluate their actions and regulate the use of external resources. However, despite growing interest in GenAI-assisted writing, SRL-informed empirical research in this area remains scarce. Existing studies are limited in both scope and methodological approach, leaving important aspects of learners' regulatory engagement with GenAI underexplored. In particular, two key gaps can be identified.

First, although extensive SRL research has shown that learners systematically differ in regulatory strategies, metacognitive monitoring, and evaluative practices across performance levels (Cheng et al., 2024; Pi et al., 2024), it remains unclear whether and how such performance-based differences are manifested in learners' interactions with generative AI during academic writing. While emerging studies acknowledge variability in GenAI use, they rarely examine how high- and low-performing writers regulate AI use in different ways.

Second, although recent research has begun to conceptualize SRL as an important explanatory framework for GenAI-assisted writing, most studies rely on retrospective methods such as questionnaires, interviews, or post-task reflections (e.g. Cui et al., 2025). While informative, these approaches provide limited insight into how regulatory behaviors unfold dynamically during writing and how learners make moment-to-moment decisions when interacting with GenAI. As a result, the behavioral mechanisms through which SRL shapes GenAI-assisted writing remain insufficiently specified.

To address these gaps, the present study adopts a process-oriented approach grounded in the SRL framework (Tang et al., 2023; Zhang & Wang, 2024). By examining high- and low-performing EFL writers, the study contributes to current research by showing how differences in writing performance are associated with differences in self-regulatory engagement with GenAI during academic writing. Rather than treating GenAI use as a uniform practice, the study highlights performance-related variation in how learners plan, monitor, and evaluate their interactions with AI support.

Specifically, the study focuses on two observable regulatory behaviors that are central to AI-assisted writing: *questioning behavior*, which reflects learners' problem identification and planning processes, and *feedback adoption behavior*, which reflects learners' evaluation, revision, or rejection of AI-generated responses. Tracing these behaviors during the writing process provides concrete, process-level evidence of how self-regulation is enacted in real time and how it differentiates more and less successful writers' use of generative AI.

From a pedagogical perspective, by highlighting performance-related variation in questioning and feedback adoption behaviors, this study offers practical insight into why learners benefit from GenAI to different extents in academic writing. The findings suggest that effective integration of GenAI requires instructional attention to how learners regulate their engagement with AI support, particularly in formulating problems and evaluating AI-generated feedback. More broadly, the study underscores the need to support less successful writers in developing more self-regulated use of generative AI, rather than assuming that access to AI tools alone will lead to effective learning.

2. Theoretical framework: self-regulated learning perspective

Writing involves recursive subprocesses of planning, translating, and reviewing (Flower & Hayes, 1981). Across these subprocesses, learners may repeatedly turn to generative AI at different moments of composing, creating multiple opportunities for AI-mediated support within the evolving writing process.

Self-regulated learning (SRL) is commonly conceptualized as a cyclical process in which learners plan, monitor, and evaluate their learning through the regulation of cognition, behavior, and external resources (Panadero et al., 2016; Zimmerman, 2002). SRL is particularly salient in complex, ill-structured tasks that require ongoing strategic decision-making (Pan et al., 2025; Wang et al., 2025), such as academic writing.

Within this context, interactions with generative AI can be viewed as localized instances of regulatory activity embedded within the writing process. Generative AI use in writing broadly aligns with Zimmerman's (2002) three-phase model—forethought, performance, and self-reflection—as illustrated in Figure 1. Rather than unfolding once in a linear sequence across the entire task, SRL in GenAI-assisted writing may occur through recurring micro-level regulatory episodes embedded within planning, drafting, and revising.

In the forethought phase, learners interpret task demands, diagnose writing difficulties, and formulate goals and help-seeking strategies. Generative AI affords low-anxiety, non-judgmental interaction and provides linguistic and reasoning scaffolding on demand (Lu et al., 2025), which may lower affective barriers to help seeking and support early problem conceptualization. However, learners differ in their ability to strategically diagnose problems and translate emerging difficulties into effective questions. Consequently, questioning behaviors at this stage can index learners' capacity for anticipatory regulation, shaping how productively they mobilize AI resources before and during text production.

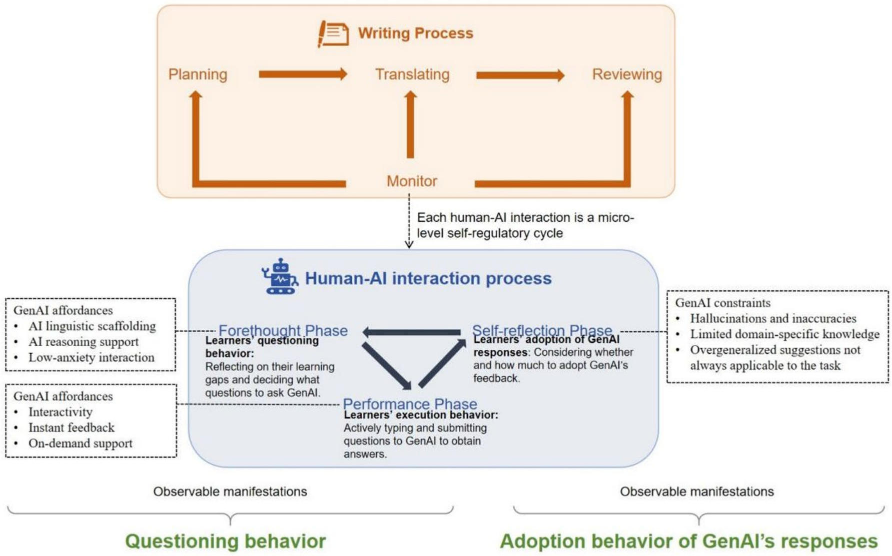

Figure 1. Self-regulatory micro-cycles embedded within GenAI-assisted academic writing.

During the performance phase, learners engage in text production while articulating questions in response to emerging difficulties that require immediate support. In this phase, generative AI affords interactivity, instant feedback, and timely linguistic or procedural assistance (Chien et al., 2025), enabling learners to seek help at the moment problems arise. Questioning behaviors therefore remain central, functioning as monitoring-oriented regulatory actions through which learners articulate emerging problems, clarify uncertainties, and regulate their engagement with AI during ongoing writing. Differences in performance may emerge not merely from access to these affordances, but from learners' differential ability to deploy questions strategically in order to align AI support with evolving writing needs.

The self-reflection phase foregrounds the evaluative demands of AI-assisted writing. Given known constraints of generative AI—such as occasional inaccuracies, limited domain specificity, and overgeneralized suggestions that may not fully align with task requirements (Alkaissi & McFarlane, 2023; Ji et al., 2022)—effective regulation at this stage requires learners to critically evaluate AI-generated output and make informed decisions about whether and how to incorporate it in relation to task goals and disciplinary expectations. Differences between high- and low-performing learners may therefore arise from variation in these evaluative judgments, rather than from differential access to AI support.

Operationally, the present study focuses on two observable manifestations of regulatory activity in GenAI-assisted writing: learners' questioning behaviors as indicators of regulatory engagement spanning the forethought and performance phases, and their adoption and revision of AI-generated responses as indicators of evaluative regulation in the self-reflection phase. From an SRL perspective, differences between high- and low-performing learners are thus expected to emerge from how effectively they leverage AI affordances to regulate planning, monitoring, and evaluation across these interconnected phases. This framework provides a theoretically grounded lens for examining how AI-mediated writing behaviors may differ across performance levels.

3. Literature review

3.1. GenAI-assisted EFL academic writing

High-quality, sustained, and process-oriented feedback is widely recognized as essential for the development of academic writing, particularly for EFL learners. In authentic instructional contexts, however, instructors often face practical constraints in providing timely and individualized feedback, resulting in delayed or surface-level responses that limit learners' opportunities for monitoring and revision (Safari & Ahmadi, 2024; Shadiev & Feng, 2024).

Against this backdrop, generative AI (GenAI) has emerged as a scalable source of writing support, capable of assisting learners across multiple stages of the writing process, including idea generation, drafting, revising, and language editing (Liu et al., 2024; Nguyen et al., 2024). Evidence suggests that AI-mediated support can improve efficiency, linguistic accuracy, and task fluency (Wang and Ren, 2024). At the same time, concerns remain regarding hallucinated content, overreliance, and potential erosion of independent writing skills (Giray, 2024; Han, 2025; Ji et al., 2022).

These mixed findings suggest that evaluating GenAI effectiveness based solely on writing outcomes is insufficient. A more pressing question concerns how learners actively engage with AI during writing and why such engagement produces differential learning outcomes. Addressing this question requires a focus on learners' regulatory processes rather than outcomes alone (Shen & Bai, 2024; Zimmerman, 2002), linking naturally to a SRL perspective.

3.2. SRL in GenAI-assisted EFL academic writing

GenAI introduces a set of technological affordances—such as accessibility, responsiveness, and interactive feedback—that require active learner regulation to become pedagogically meaningful (Wang et al., 2024b). Across digital learning contexts, research consistently shows that learners' ability to benefit from technology depends less on access itself and more on how learners regulate engagement (Jia et al., 2026; Lai et al., 2022; Lu et al., 2024).

SRL theory provides a conceptually grounded lens for understanding this variation. Research demonstrates that learners differ systematically in regulatory strategies, metacognitive monitoring, and evaluative judgment across performance levels (Cheng et al., 2024; Pi et al., 2024). This body of work suggests that learners with different levels of writing performance are also likely to interact with generative AI in distinct ways. Such performance-related differences provide a theoretically grounded basis for examining variation in how learners strategically engage with external resources, including generative AI. Applying SRL to GenAI-assisted writing suggests that learners' outcomes reflect how effectively they enact micro-level regulatory cycles when interacting with AI, rather than AI access alone.

Although SRL has been increasingly adopted as an explanatory framework in AI-assisted writing research, most studies rely on retrospective self-report methods (questionnaires, interviews, post-task reflections; e.g. Cui et al., 2025), which offer limited insight into how regulation unfolds dynamically. There is therefore a critical need for process-oriented, behavior-based analyses capable of capturing real-time regulatory behaviors in authentic writing tasks.

3.3. Behavioral dimensions of SRL in GenAI-assisted academic writing

As outlined in the theoretical framework, SRL micro-cycles recur throughout GenAI-assisted writing. To examine these micro-cycles empirically, we focus on observable behaviors that reflect learners' regulatory engagement. In particular, questioning and feedback adoption provide concrete windows into how learners plan, monitor, and evaluate their writing in interaction with AI.

Questioning behavior reflects how learners externalize writing problems and strategically seek assistance by formulating prompts to GenAI. In GenAI-assisted writing contexts, the quality of AI-generated responses is highly dependent on the clarity, specificity, and goal orientation of learners' prompts (Giray, 2023). As such, questioning behavior serves as a salient indicator of learners' problem diagnosis, goal setting, and planning processes. Feedback adoption behavior, in turn, captures how learners evaluate, revise, or reject AI-generated feedback before integrating it into their texts, thereby indexing learners' evaluative judgment and regulatory control over external input (Zou et al., 2025).

Despite growing scholarly interest in these behaviors, existing research remains limited in several respects. Studies examining questioning in AI-supported learning have largely focused on categorizing question types (e.g. Usher & Amzalag, 2025; Wang et al., 2024a), with relatively limited attention to how questioning patterns reflect learners' regulatory engagement during writing. Similarly, research on AI-generated feedback has often focused on learners' perceptions of feedback usefulness (e.g. Escalante et al., 2023; Tiwari et al., 2024), while providing limited behavioral evidence of how learners actually adopt, adapt, or reject feedback in the course of composing.

Methodologically, much of the existing literature relies on self-report data, which are vulnerable to recall bias and are unable to capture the dynamic unfolding of regulatory behaviors (Fischer et al., 2024; Wang et al., 2024a). Moreover, studies that employ objective interaction data have often been conducted in controlled learning environments such as cognitive training tasks or educational games (Elim, 2026; Chien et al., 2025), limiting their applicability to authentic EFL academic writing contexts. These limitations highlight the need for process-oriented, behavior-based analyses that examine how learners regulate their interactions with generative AI during real writing tasks.

3.4. Learners' perceptions of GenAI in academic writing

Beyond observable regulatory behaviors, learners' perceptions of GenAI constitute an important interpretive component of self-regulated engagement in academic writing. Within an SRL perspective, such perceptions can shape learners' strategic decisions about when and how to use external resources, as well as how critically to evaluate AI-generated feedback (Kim et al., 2025b).

Existing research on GenAI in education has primarily examined learners' general attitudes toward AI tools (e.g. Alzubi et al., 2025; Wang and Ren, 2024). While these studies provide useful insights into learners' acceptance of AI technologies, they often rely on survey-based measures and are not grounded in learners' experiences during specific learning tasks. As a result, it remains unclear how learners' perceptions of GenAI emerge from actual writing activities and how these perceptions relate to their regulatory behaviors during AI-assisted composing.

Moreover, although prior studies suggest that learners with different levels of performance may engage with technological support in different ways (Song et al., 2025; Wang and Ren, 2024), systematic comparisons of GenAI-related perceptions across performance levels remain limited. Understanding how high- and low-performing learners interpret the affordances and limitations of GenAI may therefore provide additional insight into why learners benefit from AI support to different extents.

3.5. The present study

In summary, although prior research has acknowledged the relevance of SRL in GenAI-assisted academic writing, several interrelated gaps remain. First, systematic comparisons of how high- and low-performing EFL learners self-regulate their use of GenAI during academic writing are still limited. Second, existing studies have rarely examined questioning behavior and feedback adoption as complementary, process-level manifestations of learners' self-regulation when interacting with GenAI. Third, much of the current evidence relies on post-task reflections, which offer limited insight into how self-regulatory behaviors unfold in real time during the writing process. In addition, learners' task-specific perceptions of GenAI in academic writing contexts have received relatively limited empirical attention, particularly across different performance levels.

To address these gaps, this study provides a process-oriented, theory-driven analysis of how learners regulate interactions with generative AI, advancing understanding of SRL in AI-assisted EFL academic writing. Anchored in the SRL framework, the study systematically compares high- and low-performing EFL learners across three dimensions: questioning behavior, adoption of GenAI-generated responses, and perceptions of GenAI in academic writing. Accordingly, the study is guided by the following research questions:

> Research question 1: Do high- and low-performing learners differ in their questioning behavior when interacting with GenAI during the academic writing process? If so, what are the differences in their behavioral patterns?
>
> Research question 2: Do high- and low-performing learners differ in their adoption behavior of GenAI's responses during the academic writing process? If so, what are the differences in their behavioral patterns?
>
> Research question 3: How do high- and low-performing learners differ in their perceptions of GenAI's strengths and limitations during the academic writing process?

4. Method

4.1. Participants

The participants were students enrolled in the *Educational Research Methods* course at a large public university in Asia. This eight-week mandatory course, part of the National Excellence Teacher Training Program, was specifically designed for senior undergraduate and first-year postgraduate students from science and engineering backgrounds who planned to pursue careers as primary or secondary school teachers. Importantly, all participants shared a highly similar educational background, disciplinary orientation, and instructional context, as they were drawn from the same cohort, completed identical course tasks, and were subject to the same instructional requirements and assessment criteria.

The course focused on foundational concepts in educational research and academic writing, and the study was integrated into the course. A convenience sampling strategy was employed, with participants drawn from students currently enrolled in the course. Prior to the start of the study, all students were informed of the research purpose and provided written informed consent. Two students who did not use GenAI tools during the writing task were excluded from the analysis, resulting in a final sample of 35 participants. Although these students were at an advanced stage of their academic programs, the majority had not received formal training in academic writing. They had not written structured literature reviews or research papers, and their writing experience was largely limited to general course essays. In addition, their exposure to academic writing in English was minimal. Therefore, they could be regarded as novice academic writers.

Each student's essay was independently evaluated by two doctoral students with demonstrated expertise in academic writing and successful publication records in English. Scoring was based on a standardized writing performance rubric (see Table 1). One rater evaluated all submissions, while the second independently assessed a randomly selected 50% subset to ensure interrater reliability. The Intraclass Correlation

Table 1. Writing scoring rubric.

Dimensions	Indicators	Explanation
Content quality (40 points)	Lead-in (10 points)	The introduction should be clear, concise, and directly relevant to the topic.
	Depth of Analysis (20 points)	The analysis should be thorough, insightful, and demonstrate critical thinking.
	Summary (10 points)	The summary should clearly address and relate to the research question, highlighting key findings.
Organization and structure (40 points)	Literature Categorization (10 points)	Literature sources should be logically grouped to support the overall argument.
	Logic flow of the text (30 points)	The text should develop clearly and logically, culminating in a well-defined research question.
Language and expression (20 points)	Accuracy (10 points)	Grammar, spelling, and punctuation should be correct, with minimal errors.
	Fluency and Professionalism (10 points)	The language should be fluent and professional, appropriate for academic writing.
Total	100 points	

Coefficient (ICC) for total scores was 0.755 (p < .001), indicating a strong level of agreement. Consensus has been reached after careful discussion. Based on the total scores, all 35 students were ranked in descending order. We drew on established precedents in educational research by adopting an upper/lower 40% split to distinguish higher- and lower-performing students. While some studies have employed median splits or equal 50/50 divisions (e.g. Echeverria et al., 2025; Pi et al., 2024), such approaches risk including learners with intermediate performance levels, thereby blurring group distinctions. To balance sample size with clearer performance differentiation, the upper/lower 40% criterion has been used in prior studies to capture more contrastive learner profiles (Chen et al., 2024; Wang & Lin, 2007).

In addition, although the grouping was not determined solely on statistical grounds, we conducted a non-parametric Mann–Whitney U test due to the small sample size to examine whether the resulting groups were empirically distinguishable. The results showed that the higher-performing group significantly outperformed the lower-performing group ($U = 0.000$, $Z = -4.509$, $p < .001$, $r = 0.85$), indicating a large effect size. This finding suggests that the grouping yielded a clear performance contrast within the present sample, supporting its suitability for the intended contrastive analysis.

To further examine group comparability, we compared students' scores on the CET-4—a nationally administered standardized English test in China. Another Mann–Whitney U test showed no significant difference between Group H (*Median* = 575.00, $SD = 47.88$) and Group L (*Median* = 545.00, $SD = 56.32$), $U = 59.00$, $Z = 0.751$, $p = 0.453$. This suggests that the two groups had comparable levels of English proficiency.

4.2. Experimental procedure

The experimental procedure is outlined in Figure 2. During Weeks 1 to 6, the course introduced essential research methods, including literature reviews, research design, data collection, and data analysis. In Week 7, students received seven articles focused on the use of gamification in synchronous video-conferencing learning. They were instructed to examine these articles closely, with attention to key elements such as research design, variables studied, measurement methods, results, and limitations. This preparatory task aimed to equip students with foundational knowledge for the formal writing assignment in Week 8.

The formal writing experiment was conducted in the final session in Week 8. Students were tasked with writing a 300-word literature review based on the seven articles. Before starting, participants completed a basic information questionnaire (covering gender, age, and CET-4 scores). Following this, they were provided with writing guidelines and a brief tutorial on using GenAI. Students had 80 min to complete the literature review assignment, with the option to interact with GenAI as needed. The entire writing process was recorded using screen capture software.

Upon completing the assignment, students submitted both their literature reviews and the chat logs documenting their interactions with GenAI. They then were invited to participate in semi-structured interviews to further explore their perceptions of GenAI.

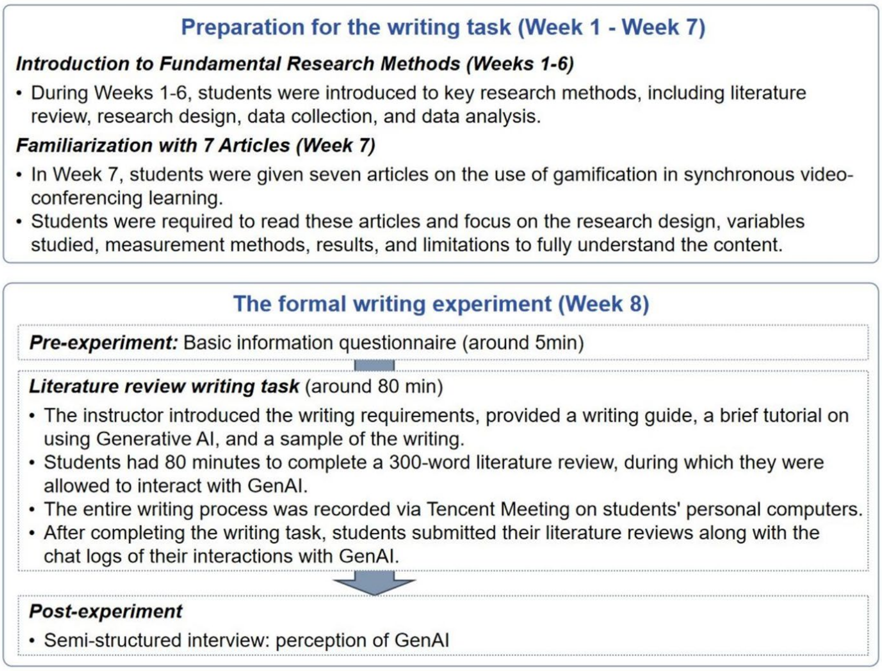

Figure 2. The experimental procedure.

4.3. Measures

4.3.1. Writing performance

When establishing assessment criteria for writing performance, we recognized the limitations of commonly used essay scoring rubrics (e.g. Ebadi & Rahimi, 2018; Teng, 2016), which are generally designed for broad essay types and lack the specificity required for academic writing. Traditional academic writing assessments, on the other hand, often evaluate full academic papers, including sections like literature review, research design, data analysis, and discussion. These comprehensive rubrics, while thorough, tend to be overly complex and not entirely suitable for evaluating the specific literature review writing task in this study.

To address these challenges, we adapted the ESL Composition Profile proposed by Jacobs et al. (1981) to create a customized scoring rubric. As illustrated in Table 1, this rubric assesses writing based on three essential dimensions: content quality, organization and structure, and language and expression, ensuring that the evaluation criteria align with the skills most pertinent to this task.

4.3.2. Perceptions of GenAI among high and low-performers

To explore differences in students' perceptions of GenAI between Group H and Group L, we conducted semi-structured interviews following the writing task. The interview protocol was informed by prior research on students' perceptions of GenAI and related technologies in writing and educational contexts (e.g. Gasaymeh et al., 2024; Yang & Lin, 2025). Guided by a two-dimensional framework—perceived advantages and perceived limitations—we designed questions to elicit both positive and critical reflections. Sample questions included: 'In what ways did GenAI help your writing process?' and 'What difficulties did you encounter while interacting with GenAI?' These types of questions are commonly employed in studies exploring learners' experiences with educational technologies (e.g. Kim et al., 2025b on GenAI use; Vaportzis et al., 2017 on tablet use) and are effective in capturing students' authentic perspectives and critical evaluations.

To encourage participation, we scheduled interviews prior to exam week, offering students a two-week window to select a convenient time. For those unable to attend during the set times, alternative arrangements were available. As a result, 11 of 14 students from Group H and 10 of 14 from Group L participated. Each interview lasted about 10 min and was conducted individually in a semi-structured format. With the students' consent, interviews were recorded and later transcribed into English for analysis. Participant identities were kept confidential throughout the process.

4.4. Data collection and data analysis

This study adopted an exploratory, process-oriented design to examine how EFL learners with different performance levels engaged with GenAI during academic writing. Analyses focused on three dimensions: questioning behavior, adoption and revision of GenAI responses, and learners' perceptions of GenAI. Table 2 summarizes the data sources and analytic methods for each research question.

To examine whether Group H and Group L differed in their questioning behavior with GenAI, we analyzed students' chat logs along two dimensions: (1) the number and types of questions, and (2) question quality.

- Number of questions: All questions were extracted and counted. Due to the small sample size ($n = 14$ per group), Mann–Whitney U tests were used to compare question quantity, which is robust under non-normality and small samples (Happ et al., 2019; McGee, 2018).
- Types of questions: An inductive, bottom-up coding approach was used to let categories emerge from the data. Two authors first jointly reviewed all chat logs, then independently conducted initial coding and discussed

Table 2. Data sources and analysis methods for each research question.

RQs	Dimensions	Data collection	Data analysis method
RQ1:Questioning behavior	-Number of questions -Types of questions -Quality of questions	Chat logs	Descriptive statistics, Mann–Whitney U test, selection of illustrative cases
RQ2:Adoption of GenAI's responses	-Utilization of GenAI's responses -Revision of GenAI's responses	Screen recordings, submitted assignments	Fisher–Freeman–Halton exact test, Fisher's exact test, qualitative content analysis, selection of illustrative cases
RQ3: Perceptions of GenAI	- GenAI's advantages and limitations	Interview	Qualitative content analysis

discrepancies to develop a framework with five categories: language-related, comprehension-related, content-related, knowledge-related, and training-related questions. An independent coder then coded the remaining data, allowing multiple codes per question when applicable. Reliability was checked by randomly double-coding 20% of the chat logs, yielding high agreement (Cohen's Kappa = 0.933). Descriptive statistics were then used to compare question type distributions between Group H and Group L.

- Quality of questions: Question quality was assessed using Jacobsen and Weber (2023) framework, based on three criteria: context, task, and clarity. Each criterion, evaluating inclusion of relevant background, task details, and clarity, was rated 0–2 points. The first author and the independent coder scored the questions, with 20% double-coded to check reliability (ICC = 0.961, $p < 0.001$). Mann–Whitney U tests compared quality scores between the two groups under small-sample conditions.
- Illustrative cases: To provide concrete examples of group-level patterns, one representative case was selected from each group to illustrate typical questioning behaviors of high- and low-performing learners.

RQ2 examined whether Group H and Group L differed in how they utilized and revised GenAI-generated content. Two dimensions were analyzed: response utilization and response revision. Response utilization refers to the extent to which students incorporated AI content, categorized as full, partial, or non-utilization. Response revision indicates whether utilized content was altered, classified as revised or not revised.

Students' utilization and revision behaviors were assessed through analysis of screen recordings and submitted assignments. Details are shown in Table 3. Due to limitations in tracing AI responses to 'understanding' questions (47 chat logs) and missing screen recordings for six students (2 from Group H, 4 from Group L), analyses focused on content-, language-, knowledge-, and training-related questions, totaling 107 chat logs. Inter-coder reliability was verified by randomly double-coding 20% of the data, resulting in Cohen's Kappa values of 0.958 for utilization and 0.837 for revision, indicating high agreement. Given the categorical data and small sample, group comparisons used exact tests: Fisher's exact test for 2×2 tables and Fisher–Freeman–Halton exact test for larger tables (Cochran, 1954). All tests were two-sided, with effect sizes reported where applicable.

Among instances in which revision occurred, we conducted a more fine-grained classification of revision types. Given the limited number of revision instances, analyses at this level were descriptive rather than inferential. To guide the classification, we drew on Faigley and Witte (1981) revision framework. Because the present study focused specifically on learners' revisions of AI-generated responses rather than on global essay restructuring, we retained three levels of revision reflecting increasing cognitive engagement—formal changes, meaning-preserving changes, and microstructure changes—while excluding macrostructure changes, which involve broader organizational restructuring beyond individual AI outputs: (1) *Formal changes:* Surface-level edits (grammar, spelling, punctuation, tense, minor lexical adjustments) that improve accuracy without altering meaning. (2) *Meaning-preserving changes:* Paraphrasing or rewording that maintains propositional content while

Table 3. Criteria for determining response utilization and response revision.

Dimension	Explanation	Example
Response utilization		
Full utilization	Using all the provided content.	• Example 1:From 01:15:39 to 01:15:59, Student N typed and sent the following message in the GenAI chat box: • '*Please help me polish the following text in English: The outbreak of COVID-19 has led many schools to adopt online synchronous video teaching, and gamification applications have been widely studied as a method to improve online learning engagement and learning outcomes…*'From 01:15:59 to 01:16:07, GenAI responded with: • '*The COVID-19 pandemic has prompted numerous schools to embrace online synchronous video teaching. In response, gamification applications have garnered significant attention as a strategy to enhance engagement and improve learning outcomes in the online learning environment…*'From 01:16:07 to 01:16:38, Student N copied the entire GenAI response into their draft.
Partial utilization	Using some of the provided content.	• Example 2: • From 01:16:38 to 01:17:06, Student H copied the first paragraph of their draft into the GenAI chat box and issued the instruction:'*A growing body of research has examined the use of gamification in videoconferencing-supported classes. An in-depth review of these studies is beyond the scope of this article. Here, we briefly describe some studies from the past three years. Please, as an education master's student, complete a literature review using academic language and appropriate wording to rephrase the above sentences.*'From 01:13:50 to 01:14:07, GenAI responded with: • '*A burgeoning body of scholarship has scrutinized the integration of gamification within the framework of videoconferencing-supported classes. Given the voluminous nature of this research, a comprehensive examination exceeds the confines of this article. Consequently, we provide a succinct overview of select studies conducted in the past three years.*'From 01:16:38 to 01:17:06, Student H only copied the final sentence from the GenAI response: '*Consequently, we provide a succinct overview of select studies conducted in the past three years.*' into their draft.
Non-utilization	Not using any of the provided content.	• From 00:17:32 to 00:19:28, Student K typed and sent the following message in the GenAI chat box:'*Please help me draft an outline for this literature review. It should reflect changes in research content and highlight the main viewpoints and important experiments in this field. Additionally, ensure the outline is both academic and rigorous, with some creative elements.*'From 00:19:28 to 00:19:42, GenAI generated a response. According to the screen recording, Student K did not adopt the GenAI response into their draft.

(*Continued*)

Table 3. Continued.

Dimension	Explanation	Example
Response revision		
No revision	Using the content as is.	Both Example 1 and Example 2 fall under 'no revision'.
Revision	Making changes to the provided content before using it.	• Example 4:From 00:46:55 to 00:47:17, Student Q typed and sent the following message in the GenAI chat box: • *'Translate the following into English: In studies conducted over the past three years, one category is experimental research. Most experimental studies confirm that gamification technology has a positive impact on conducting synchronous video conference classrooms.'*From 00:47:17 to 00:47:21, GenAI responded: • *'In the research conducted in the past three years, one category is experimental research. The majority of experimental studies confirm that the application of gamification technology has a positive facilitating effect on the conduct of synchronous video conference classrooms.'*From 00:47:21 to 00:48:22, Student Q copied GenAI's response into their draft and then revised 'research' to 'study' and 'video conference classrooms' to 'online learning'.

improving clarity, conciseness, style, or contextual appropriateness, indicating moderate evaluative engagement. (3) *Microstructure changes:* Local revisions affecting cohesion, logical flow, or rhetorical clarity, reflecting higher-level cognitive engagement with the communicative adequacy of AI output.

Single revision instances could be coded across multiple categories when serving multiple functions. Inter-coder agreement for this classification was high (Cohen's Kappa = 0.839), demonstrating reliable application of the coding scheme.

To more vividly illustrate the differences in utilization and revision patterns between high- and low-performing learners, this study selected illustrative cases for in-depth analysis, showcasing the typical behavioral characteristics of the two groups when processing GenAI-generated responses.

To address RQ3, which examines differences in perceptions of GenAI between Group H and Group L, data were obtained from the interviews. Then, we used a bottom-up inductive approach. Responses related to students' GenAI perception were coded into thematic units, which were then organized into potential themes. To ensure the reliability of the analysis, two independent coders re-coded the entire dataset. Cohen's Kappa test indicated high reliability with a score of 0.918. Any discrepancies between the coders were resolved through discussion.

5. Results

5.1. RQ1: Questioning behavior

To address RQ1, which investigates whether high-performing and low-performing students differ in their questioning behavior when interacting with GenAI, we conducted a comparative analysis along two dimensions: the number and types of questions posed, and the quality of those questions.

5.1.1. Number and type of questions

Table 4 summarizes the number of questions posed by Group H and Group L. On average, Group H asked 5.71 questions, slightly fewer than Group L's 7.00. A Mann-Whitney *U* test indicated no significant difference in question quantity between Group H and Group L, $U = 96.500$, $z = -0.069$, $p = 0.945$, $r = -0.013$, 95% CI [−0.39, 0.36], suggesting that the frequency of questions posed during GenAI interaction was comparable between high- and low-performing students.

Beyond quantity, we analyzed the types of questions using an inductive coding approach (see Section 2.4). This analysis identified five thematic dimensions: *Language*, *Understanding*, *Content*, *Knowledge*, and *Training* (see Table 5).

In the *Language* dimension, Group H posed more questions (Group H: 47; Group L: 31), with a particularly notable difference in the *polishing writing* category (Group H: 24; Group L: 7), indicating that high-performing students were more focused on refining language and expression using GenAI.

In the *Understanding* dimension, Group L raised more questions (Group H: 15; Group L: 33). The most substantial difference was observed in the *translating reading materials into Chinese* category (Group H: 8; Group L: 20), suggesting that low-performing students relied more heavily on GenAI for translation and comprehension tasks. Group L also used GenAI more frequently for *summarizing and categorizing literature* (Group H: 2; Group L: 9), further reflecting their dependence on AI for processing source materials.

In the *Content*, *Knowledge*, and *Training* dimensions, Group L also posed slightly more questions (*Content*: Group H: 14; Group L: 20; *Knowledge*: Group H: 7; Group L: 12; *Training*: Group H: 10; Group L: 18), though the differences were not pronounced and are thus not discussed in detail.

Overall, high-performing students tended to focus on language-related questions, particularly text polishing, while low-performing students made greater use of GenAI for comprehension support and content generation.

5.1.2. Illustrative typical cases of questioning behavior

To concretely illustrate the group-level patterns identified above, we present two representative cases drawn from Group L (Student M) and Group H (Student J), whose questioning behaviors and post-hoc reflections exemplify divergent orientations toward GenAI use.

Case 1: Student M (Group L)—Generative Content-Oriented Use

Student M posed multiple questions ($n = 5$), frequently requesting GenAI to generate extended texts and iteratively develop literature review drafts. His prompts repeatedly asked the AI to synthesize multiple sources, expand background information, and produce upgraded versions of drafts, indicating a stronger reliance on GenAI for content generation and macro-level development.

Table 4. The number of questions between Group H and Group L.

	N	*Min*	*Max*	*Mean*	*S. D.*	*Median*
Group H	14	1	14	5.71	3.00	6.00
Group L	14	1	35	7.00	8.55	5.50

Table 5. Differences in question types between Group H and Group L.

Category	Dimensions	Frequency		Examples
		Group H	Group L	
Language	Total	47	31	• 'Please translate the following mixed Chinese-English draft entirely into English.' • 'Can you polish the following passage for me…?'
	Translating students' draft to English	23	24	
	Polishing writing	24	7	
Understanding	Total	15	33	• 'Please provide an overview of this reading material.' • 'Translate this section of the reading material into Chinese…'
	Translating reading materials to Chinese	8	20	
	Summarizing and categorizing literature	2	9	
	Translating GPT's responses to Chinese	3	3	
	Grasping the gist of reading materials	2	1	
Content	Total	14	20	• 'Revise and expand…' • 'Combine the following two sections into one paragraph…' • 'Please help me improve the logic and make the structure more complete.'
	Providing ideas for writing	7	4	
	Generating content from simple instructions	5	7	
	Summarizing content of the draft	0	4	
	Structure and Logic of the draft	2	0	
	Expanding content of the draft	0	1	
	Referencing styles for citations	0	4	
Knowledge	Total	7	12	• 'Tell me how to write a literature review.' • 'Specifically, the concept of gamification includes three major elements. Please describe these three elements.'
	Domain knowledge, such as inquiries about specific terms	7	11	
	Genre knowledge, such as how to write a literature review	0	1	
Training	Total	10	18	'Please act as an expert in the field of education, review the following literature review, and provide suggestions…'
	Refining the output of GenAI by revising prompts or incorporating additional contextual information to generate responses that more effectively align with their needs or expectations.			

Prompt 1: Based on the provided literature, please compose a literature review of approximately 300 words in English.

Prompt 2: Provide an upgraded version that includes the background, contributions, and limitations of these studies, and leads into the research question.

Prompt 3: Please generate another upgraded version, this time also incorporating the following two additional articles: [Article A] and [Article B].

Prompt 4: Draft a literature review that adheres to standard academic writing conventions for such sections.

In the interview, Student M framed GenAI primarily in terms of efficiency and reduced effort during writing:

> It's very efficient. You don't need to struggle over wording. You just give it your draft, and it checks common mistakes and makes the text more fluent.

Case 2: Student J (Group H)—Strategic Language-Oriented Use

In contrast to Student M's content-generative orientation, Student J's questioning behavior reflected a strategic focus on linguistic execution rather than content construction. Student J posed four questions, primarily using GenAI for translation and linguistic polishing while retaining full control over content and overall structure. His prompts consistently targeted sentence-level refinement of pre-written text, suggesting a delegation of linguistic execution rather than conceptual work.

> Prompt 1: ……As a master's student in education, please translate the above content into academic English.

> Prompt 2/3: ……Please translate the above content into academic English.

> Prompt 4: ……Please replace the above statement with suitable academic language.

Interview data further support this interpretation. Student J emphasized maintaining authorial control over the intellectual organization of the literature review, stating:

> I still want to structure the whole paper based on my own ideas. I mainly use GPT for translating from Chinese to English or for small-scale polishing.

5.1.3. Quality of questions

Descriptive statistics for overall question quality and its sub-dimensions are reported in Table 6. No reliable difference was observed in overall question quality between Group H and Group L, $U = 3808.000$, $z = -0.366$, $p = 0.714$, $r = -0.027$, 95% CI [−0.17, 0.12], indicating a negligible effect size and minimal practical difference between the two groups.

Consistent with this pattern, comparisons across the three sub-dimensions of question quality—context, mission, and clarity—also revealed no statistically reliable group differences, with all effect sizes in the small-to-moderate range (r_1 = −0.129; r_2 = −0.045; r_3 = −0.031) and p values well above conventional significance thresholds (p_1 = 0.086; p_2 = 0.549; p_3 = 0.680). After applying a Bonferroni correction for multiple comparisons (adjusted $\alpha = 0.0125$), none of these differences reached statistical significance, confirming that the lack of group differences is robust.

Table 6. Differences in the quality of questions between Group H and Group L.

	Group H			Group L		
	Mean	*SD*	*Median*	*Mean*	*SD*	*Median*
Quality of questions	4.49	1.15	5.00	4.48	1.30	5.00
Context dimension	0.95	0.48	1.00	1.07	0.46	1.00
Mission dimension	1.72	0.53	2.00	1.64	0.65	2.00
Clarity dimension	1.81	0.48	2.00	1.79	0.50	2.00

These results suggest that, in terms of question quality, high- and low-performing learners produced broadly comparable prompts when interacting with GenAI.

5.2. RQ2: Adoption behavior

5.2.1. Utilization of GenAI's responses

As described in *Section 2.4*, screen recordings were analyzed to classify students' utilization of GenAI responses as Full Utilization, Partial Utilization, or Non-Utilization (Table 3). To examine potential group differences between Group H and Group L, a Fisher–Freeman–Halton exact test was conducted (Table 7).

The results indicated no statistically reliable difference in utilization patterns between the two groups (two-sided Fisher–Freeman–Halton exact test, $p = 0.461$, *Cramér's* $V = 0.121$), suggesting broadly comparable tendencies in the overall extent to which high- and low-performing students adopted AI-generated content. To account for multiple comparisons within Section 5.2, a Bonferroni correction was applied (adjusted $\alpha = 0.025$). After correction, the group difference remained non-significant.

5.2.2. Revision of GenAI's responses

Further analyses examined whether students revised GenAI-generated responses during the writing process. Based on screen recordings, behaviors were classified as No Revision or Revision, and group differences were assessed using Fisher's exact test. The results revealed a clear group difference (two-sided Fisher's exact test, $p = 0.005$), with Group H students being significantly more likely to revise and integrate GenAI outputs into their writing than Group L students (no revision: Group $H = 19$, Group $L = 27$; revision: Group $H = 23$, Group $L = 8$).

To account for multiple comparisons, a Bonferroni correction was applied (adjusted $\alpha = 0.025$), and the group difference remained statistically significant. The corresponding odds ratio ($OR = 4.09$, 95% CI [1.51, 11.05]) indicates a meaningful difference in revision behavior between the two groups. Given the small sample size and the exploratory nature of the study, this finding should be interpreted with caution. Nevertheless, the robustness of the effect under an exact testing framework suggests that revision behavior may serve as a sensitive, process-level indicator of differential cognitive engagement in GenAI-supported academic writing.

Given the observed group differences in overall revision behavior, we further examined how students revised GenAI-generated content. Analyses at this level were restricted to instances in which revision occurred. We applied the adapted revision framework proposed by Faigley and Witte (1981), as described in Section 4.4. Due to the limited number of cases, revision behaviors were categorized into three types

Table 7. Results of exact test on utilization of GenAI's responses.

Category	Group H	Group L	*Exact p*	*Cramér's V*
Full Utilization	34	26	0.461	0.121
Partial Utilization	8	9		
No Utilization	13	17		

for descriptive analysis only, and no inferential statistical tests were conducted at this level.

Table 8 presents the frequency and representative examples of revision types across the two groups. Overall, Group H produced substantially more revisions ($n = 28$) than Group L ($n = 8$). Moreover, the distribution of revision types differed between the groups. Group H produced revisions across all three categories, including 10 formal changes, 9 meaning-preserving changes, and 9 microstructure changes, indicating engagement with multiple levels of text modification. In contrast, Group L revisions were concentrated primarily in formal ($n = 3$) and meaning-preserving changes ($n = 4$), with only one instance of microstructure-level revision.

Taken together, these descriptive patterns suggest that Group H students engaged in more frequent and varied revisions of AI-generated content, whereas Group L students made fewer revisions overall and relied primarily on surface-level edits or direct adoption of GenAI output.

5.2.3. Illustrative typical cases of adoption behavior

To provide a more concrete illustration of the group-level differences observed in revision behaviors, two representative cases were selected from the screen-recording data during the polishing stage of writing to serve as illustrative, non-exhaustive examples of the observed adoption patterns.

Table 8. Types of revisions applied to GenAI-generated content (adapted from Faigley & Witte, 1981).

Revision type	Examples	Group H	Group L
Formal changes	1:03:42 – Student G changed 'evaluating students' engagement' to 'evaluated students' engagement' to maintain tense consistency. 1:09:42 – Student K changed 'However, this is just an anecdotal study regarding students' perceptions of their learning experiences' to 'However, they are just anecdotal studies regarding students' perceptions of their learning experiences' to correct number agreement and align with preceding context, ensuring grammatical consistency without changing the meaning.	10	3
Meaning-preserving changes	0:51:16 – Student I changed 'the gamification applications in synchronous video conference classrooms' to 'gamification in synchronous video-conferencing-supported classes' to improve conciseness and academic phrasing without altering the propositional content. 0:42:49 – Student C changed 'gamified online learning' to 'online gamification flipped learning' to enhance clarity and precision while retaining meaning.	9	4
Microstructure changes	1:02:07 – Student C added 'Additionally' between two sentences to improve cohesion and logical flow: 'Moreover, these studies do not delve into the specifics of participation quality in the videoconferencing-supported classes' → 'Additionally, the majority of research has primarily focused on students' academic performance, neglecting other aspects of participation quality.' 0:53:45 – Student A expanded a statement to improve clarity and completeness: '…but lacks the impact on the quantity and quality of student participation' → '…but lacks the impact on the quantity and quality of student participation and lacks an objective and accurate assessment method.'	9	1

A representative lower-performing student (Student S) from Group L tended to adopt GenAI-generated responses with minimal or no revision. As illustrated in Figure 3, Student S translated and inserted GenAI output into the draft largely verbatim, without intervening at the lexical, rhetorical, or structural level. Similar instances of direct adoption were observed at multiple points during the writing session; Figure 3 presents one representative episode of this pattern.

By contrast, a representative higher-performing student (Student C) from Group H actively revised GenAI-generated content before integrating it into the draft. As shown in Figure 4, Student C replaced general expressions with more discipline-appropriate terminology using the coding book, improved sentence-level cohesion through the addition of transitional markers, removed awkward or redundant phrasing, and inserted relevant citations. This pattern of critically translating and revising GenAI output occurred consistently throughout the writing process; Figure 4 illustrates one such episode.

Together, these two cases illustrate the different orientations toward GenAI use identified in the process-level analysis: higher-performing learners tended to engage

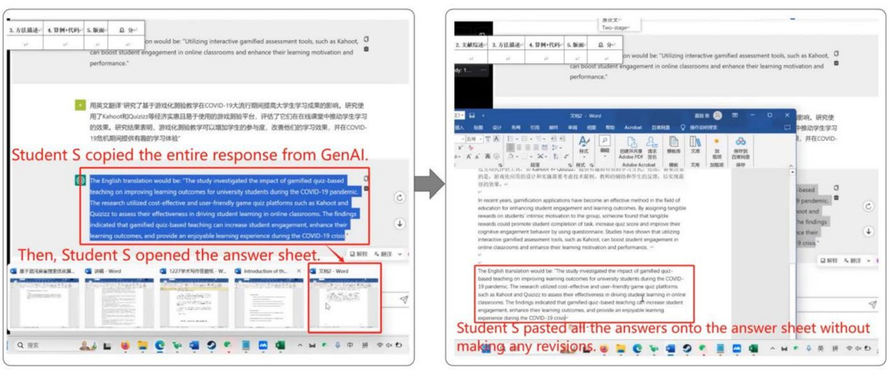

Figure 3. Illustrative snapshots of GenAI adoption during the polishing stage: a lower-performing student adopting GenAI output with minimal revision.

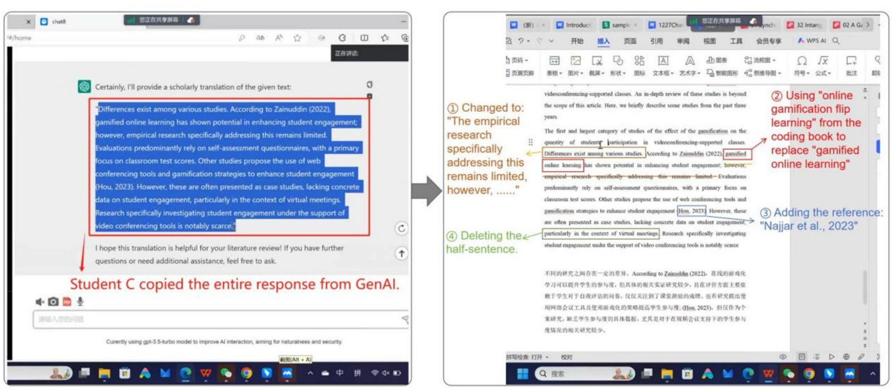

Figure 4. Illustrative snapshots of GenAI adoption during the polishing stage: a higher-performing student actively revising GenAI-generated content before integration.

in evaluative and revision-oriented integration of AI output, whereas lower-performing learners more often treated GenAI responses as ready-to-use text segments.

5.3. RQ3: Perceptions of GenAI's advantages and limitations

To answer RQ 3, which investigates whether high and low performers differ in their perceptions of GenAI's strengths and limitations during the academic writing process, we conducted interviews after the writing experiment. 11 students from Group H and 10 students from Group L participated in interviews. During the discussion on the benefits of GPT, four key themes were identified. When it came to the limitations, three key areas were highlighted. Table 9 summarizes these themes along with the frequency of mentions by students in each group.

5.3.1. Advantages of GenAI perceived by group H and group L

When discussing the benefits of using GenAI for writing English literature reviews, there was no significant difference between Group H and Group L. For both groups, the *Language* dimension was the most frequently mentioned, underscoring students' strong appreciation for GenAI's language-related functions. Specifically, all 11 students in Group H and 8 out of 10 students in Group L highlighted the Language dimension. Examples of student comments include:

> It feels quite satisfying when it polishes my writing. (Student B from Group H)

> After writing in English, I felt that the expression wasn't fluent enough, so I asked GenAI to help polish it. (Student J from Group L)

The next most frequently mentioned themes were *Content, Knowledge*, and *Understanding*. For Content, 5 out of 11 students from Group H and 6 out of 10 students from Group L acknowledged GenAI's role in supporting content-related aspects. Regarding *Knowledge*, three students from Group L and two from Group H acknowledged GenAI's role. *Understanding* received the least attention, mentioned by only one student from Group L, with none from Group H referring to it.

5.3.2. Limitations of GenAI perceived by group H and group L

When discussing the limitations of using GenAI to assist with academic writing, notable differences emerged between the perceptions of Group H and Group L. User query issues were the most frequently mentioned limitation by Group H, cited by 7 out of 11 students, compared to only 2 out of 10 students in Group L. User query refers to the ability to provide clear and effective prompts to elicit desired outputs from GenAI. For example:

> If you don't provide clear instructions, it may not produce the results you want, so you need to learn how to give proper prompts. (Student I from Group H)

In contrast, GenAI output quality was the most frequently mentioned limitation by Group L, with 8 out of 10 learners expressing dissatisfaction, compared to 6 out of 11 learners in Group H. Students highlighted issues such as inaccurate

Table 9. Advantages and limitations of GenAI perceived by Group H and Group L.

No.	Dimension	Frequency	
		Group H	Group L
Advantages			
1	Language	11	8
2	Content	5	6
3	Knowledge	2	3
4	Understanding	0	1
Limitations			
1	User query	7	2
2	GenAI output quality	6	8
3	Technical limitations	2	6

understanding by GenAI, lack of depth in responses, and lack of domain-specific knowledge. Examples include:

> Its understanding of contextual connections is still inadequate. It struggles to connect the understanding between one question and the next. (Student P from Group L)

> For certain specific fields, it seems like it doesn't have much specialized knowledge. (Student R from Group L)

Similarly, in the technical limitations dimension, Group L (6 out of 10 students) identified this issue more frequently than Group H (2 out of 11 students). Specific technical limitations mentioned by students included the text input limit, outdated database, and low credibility of responses. Examples include:

> There's a limit to the number of words it can process. (Student L from Group L)

> It can only provide data up to September 1, 2021, but nothing beyond that. (Student H from Group H)

> Sometimes it makes things up when it doesn't know the answer, which lowers my trust in it. (Student M from Group L)

Overall, both groups demonstrated a comparable appreciation of GenAI's strengths, especially its ability to enhance language expression. However, their views on its limitations differed markedly: high-performing students emphasized refining their prompts to improve output quality, while low-performing students were more concerned with the accuracy of GenAI's responses and its technical limitations.

6. Discussion

Situated within an authentic classroom context, this study examined how EFL learners engaged with GenAI during an academic writing task and how such engagement differed between higher- and lower-performing writers. By dividing learners into Group H (top 40%) and Group L (bottom 40%) based on writing performance, the analysis identified process-level differences associated with writing performance, particularly in learners' questioning, revision, and attributional behaviors.

6.1. Differential questioning orientations in AI-assisted writing

Learners in both groups engaged with GenAI selectively and at comparable frequencies, yet their questioning patterns revealed fundamentally different orientations toward AI-supported writing. Rather than differing in how often GenAI was used, high- and low-performing learners differed in the functional roles assigned to the tool during the writing process.

High-performing learners (Group H) primarily used GenAI for language-focused refinement, including translation adjustment and academic expression polishing. In contrast, low-performing learners (Group L) more frequently relied on GenAI for both content generation and comprehension support, including AI-generated paragraph drafts, Chinese translations of source materials, and literature summarization tasks. Interview data further confirmed these differences, indicating that GenAI functioned primarily as a linguistic refinement resource for high-performing learners but as a comprehension and content-support tool for lower-performing learners.

Taken together, these patterns suggest that performance differences were reflected not in the intensity of AI use but in the ways learners integrated GenAI into the writing process.

These differences cannot be attributed to general language proficiency, as CET-4 scores did not differ significantly between groups (Taladngoen et al., 2020). Instead, they reflect learners' strategic deployment of AI support during the writing process.

Drawing on Flower and Hayes (1981) cognitive writing model, high-performing learners' language-focused questions align with the reviewing stage, where attention shifts to expression and refinement. By contrast, low-performing learners' comprehension-oriented and content-generation requests correspond to earlier planning and translating stages—the former involving interpretation of source materials and task understanding, the latter involving the conversion of ideas into text. This stage-specific distribution suggests that high-performing learners tended to construct meaning independently before seeking AI assistance, whereas low-performing learners relied on GenAI to alleviate immediate production demands—processes central to the development of academic writing competence (Barrot, 2025).

These patterns also intersect with GenAI's known capability boundaries. Prior research indicates that GenAI performs relatively reliably in language polishing tasks (Kohnke, 2024; Marzuki et al., 2023; Nguyen et al., 2024), whereas content generation carries greater risks of inaccuracies and hallucinations (Alkaissi & McFarlane, 2023; Choudhuri et al., 2024). High-performing learners thus tended to engage GenAI within its areas of strength, while low-performing learners' reliance on content generation increased potential risk.

These findings contribute to ongoing debates in CALL and AI pedagogy regarding whether generative AI promotes learner dependency or supports strategic engagement. While concerns have been raised that AI may substitute core writing processes (Han, 2025; Tang, 2025), the present results suggest a more differentiated pattern in academic writing contexts. AI use was not uniformly extensive; rather, differences emerged in how strategically learners integrated GenAI into specific phases of writing. This indicates that dependency risks are not inherent to the technology itself, but are shaped by learners' regulatory capacities and phase-sensitive use.

Instruction should guide learners to pose stage-appropriate questions, emphasizing language refinement during reviewing while encouraging independent content construction and deep engagement with source materials during earlier planning stages. Such guidance can help ensure that AI support complements rather than substitutes core writing processes.

6.2. Differential revision orientations in AI-assisted writing

Beyond questioning behaviors, learners also differed in how they adopted and revised GenAI output, which can be understood as an observable manifestation of the self-reflection phase in human–AI self-regulatory micro-cycles (Zimmerman, 2002). While no significant group differences were found in adoption types (full, partial, or rejected), clear divergences emerged in the depth and nature of revisions.

High-performing learners (Group H) engaged in revisions across formal, meaning-preserving, and microstructure levels, with a notable proportion of micro-structure changes targeting cohesion, logical flow, and rhetorical clarity. In contrast, low-performing learners (Group L) focused primarily on surface-level revisions, including formal and meaning-preserving changes, with relatively few microstructure revisions (Faigley & Witte, 1981).

Taken together, these patterns suggest that performance differences were reflected not in how frequently learners adopted AI output, but in how actively they transformed it during the revision process.

From a self-regulated learning perspective, these patterns reflect different orientations toward AI support. High-performing learners maintained a critical stance, engaging in sustained monitoring and substantive revision that reinforced authorial agency. Low-performing learners, by contrast, adopted a more reliance-oriented approach, using GenAI primarily to reduce immediate cognitive load rather than to engage in evaluative, self-regulatory processing (Gao et al., 2023; Hong et al., 2025).

These findings align with prior research showing that high-performing writers demonstrate stronger metacognitive control and more frequent strategy use across contexts (Rahmawati et al., 2019; Zhang et al., 2014), and illustrate how such regulatory differences are enacted in AI-assisted academic writing.

These findings contribute to current discussions in CALL and AI pedagogy concerning the role of AI-generated feedback in writing development. While AI tools are often assumed to facilitate revision and enhance writing quality (Tran, 2025; Zhao, 2025), the present results suggest that improvement depends less on access to AI suggestions than on learners' regulatory engagement with those suggestions. AI-supported revision does not automatically lead to deeper text improvement; rather, the learning value of AI output depends on whether learners critically evaluate and transform it. Without such regulatory engagement, reliance on AI may also introduce risks, including inaccuracies, hallucinations, and potential academic misconduct (Khalil & Er, 2023; Williams, 2024), while potentially constraining the development of independent writing skills (Giray, 2024).

The findings suggest that instruction should explicitly structure revision activities to promote active transformation of AI-generated text rather than passive adoption. For example, learners may be asked to justify revisions, compare AI-generated and self-revised versions, or explain why specific AI suggestions are accepted or revised. Such practices help position revision as a regulatory process and support the development of authorial control in AI-assisted writing.

6.3. Differential attributional orientations in AI-assisted writing

Learners' perceptions of GenAI's strengths and limitations functioned as an interpretive lens shaping how they regulated AI use during academic writing. Across performance groups, learners reported largely convergent views of GenAI's strengths, consistently identifying language editing and polishing as its most valuable affordances and rating these as more critical than content generation. This convergence suggests that group differences in AI-supported writing outcomes are unlikely to stem from divergent evaluations of AI strengths.

By contrast, clear group differences emerged in how learners interpreted AI limitations. High-performing learners tended to attribute less effective outputs to inadequately formulated prompts or suboptimal interaction strategies. Low-performing learners, however, more frequently emphasized technical constraints of the system itself.

Taken together, these patterns suggest that performance differences were reflected not only in how learners used GenAI, but also in how they interpreted and responded to its limitations during the writing process.

The shared prioritization of language-related support contrasts with findings from Liu et al. (2024), where learners valued content generation in more creative or argumentative tasks. In academic EFL writing, however, argument development and source-based meaning construction are core requirements (Hirvela & Du, 2013). Against this backdrop, learners' convergence suggests that GenAI is perceived primarily as a linguistic support tool rather than a content-producing agent.

Group differences in perceptions of limitations reflect distinct attributional orientations. Attribution theory (Li et al., 2022; Weiner, 1979) suggests that attributing difficulties to internal, controllable factors supports adaptive regulation and strategy adjustment, whereas external, uncontrollable attributions may constrain reflective engagement. In this study, high-performing learners' emphasis on improving questioning strategies indicates a stronger sense of agency and a view of AI interaction as a learnable skill. In contrast, low-performing learners' tendency to externalize AI shortcomings may reduce motivation to refine interaction strategies, thereby limiting opportunities for regulatory development.

These findings further inform ongoing discussions about the nature of AI literacy in language education. Existing work often emphasizes technical skills such as prompt formulation (eg., Park & Choo, 2025) or knowledge of system functions (eg., Kim et al., 2025a). However, the present results suggest that effective AI use also depends on learners' interpretive and regulatory orientations toward AI output and limitations. From this perspective, AI literacy involves not only operational competence but also the ability to interpret and respond adaptively to imperfect AI output.

The findings suggest that instruction should address not only what GenAI can do, but also how learners interpret its limitations. Helping learners view suboptimal AI outputs as partly strategy-dependent, rather than purely technical failures, may foster stronger agency and more adaptive regulation in AI-assisted writing.

6.4. Implications

Building on the analyses in Sections 6.1–6.3, the findings collectively suggest a more differentiated understanding of effective AI-assisted academic writing. Rather than treating GenAI as a general-purpose writing aid, effective AI use emerges as a phase-sensitive and self-regulated process shaped by learners' questioning orientations, revision practices, and interpretations of AI limitations.

The following three principle-based implications synthesize these findings and translate them into a coherent pedagogical framework for AI-mediated EFL academic writing.

Principle 1: Aligning GenAI use with phases of the cognitive writing process. The findings indicate that the pedagogical value of GenAI is phase-sensitive rather than uniform across the writing process. High-performing learners primarily engaged GenAI during the reviewing or self-reflection phase to refine language, whereas low-performing learners relied on it during the translating phase for content generation and comprehension support. Use during the reviewing phase preserves conceptual engagement while reducing linguistic load, whereas translating-phase reliance may replace core cognitive processes involved in meaning construction.

Pedagogically, instructors should provide phase-specific guidance for AI use. Rather than offering general advice about GenAI, instruction should emphasize independent content development during translating and strategic AI support for language refinement during reviewing.

Principle 2: Positioning revision of AI output as a core self-regulatory competence. Although both groups adopted similar types of AI feedback, high-performing learners engaged in deeper evaluative and microstructure-level revisions. This suggests that effective AI-supported writing depends less on access to AI output than on learners' ability to monitor, evaluate, and transform it.

Instruction should therefore externalize learners' regulatory processes. Reflective prompts, justification-based revisions, and version comparison tasks can encourage students to explain why and how AI output is revised. Such practices position AI literacy as a form of metacognitive control over external input, rather than mere proficiency in prompt formulation.

Principle 3: Addressing attributional patterns in AI-mediated writing. Learners' interpretations of AI-related difficulties shaped their regulatory behavior. High-performing learners tended to attribute suboptimal outputs to controllable, strategy-related factors, whereas low-performing learners more often attributed them to fixed technical limitations. According to attribution theory (Weiner, 1979), internal, controllable attributions are more likely to support adaptive regulation.

Pedagogically, instructors should help learners reconceptualize AI limitations as partially strategy-dependent and improvable. Framing AI-related problems as opportunities to refine questioning and interaction strategies may strengthen learners' sense of agency and promote more sustained, strategic engagement with AI-supported writing.

6.5. Limitation and future work

This study has several limitations that should be considered when interpreting the findings. First, the sample size was relatively small and homogeneous, with participants divided into higher- and lower-performing groups based on relative rankings within the same cohort ($n = 14$ per group). While this classification enabled exploratory comparisons of GenAI use patterns, the limited sample size constrained statistical power and restricted the strength and generalizability of group-based inferences. As a result, the findings of this study should be interpreted cautiously, and their generalizability beyond the present sample remains limited. Moreover, grouping learners by performance inevitably abstracts away individual variation within each group. Although clear group-level tendencies were observed, learners within the same performance group demonstrated varied patterns of GenAI use. The reported differences should therefore be interpreted as general tendencies rather than uniform behavioral patterns.

Second, the study was conducted using a single, time-limited academic writing task within a specific instructional context. Although this design allowed for fine-grained analysis of students' interactions with GenAI during EFL academic writing (Jia & Lu, 2026), it does not capture how such engagement may evolve across different writing genres, extended durations, or repeated instructional cycles. As a result, the ecological validity of the findings is constrained by the one-off nature of the task.

Third, students' engagement with GenAI may have been influenced by novelty effects and individual differences in attitudes toward AI. Given that many participants had limited prior experience with GenAI-assisted writing, initial curiosity, uncertainty, or overreliance may have shaped their interaction behaviors. In addition, learners' pre-existing beliefs about AI—such as trust in its accuracy or perceptions of its role in academic work—may have acted as confounding factors affecting both adoption behavior and writing performance.

Taken together, these limitations suggest that the findings should be viewed as exploratory and context-sensitive rather than definitive. Future research should employ larger and more diverse samples, incorporate longitudinal or multi-task designs, and systematically examine learners' prior experiences and attitudes toward AI to strengthen the robustness and ecological validity of conclusions regarding GenAI-assisted EFL academic writing.

7. Conclusion

As GenAI tools become increasingly integrated into language education, understanding how learners engage with them in academic writing is crucial. Framed by a self-regulated learning perspective, this study examined high- and low-performing EFL learners' questioning behaviors, feedback adoption patterns, and perceptions of GenAI.

Findings indicate that differences between groups were qualitative rather than quantitative. High-performing learners asked primarily language-focused questions, engaged in more extensive microstructure-level revisions, and attributed AI limitations to their own prompting strategies. In contrast, low-performing learners relied more on GenAI for content generation and comprehension support, focused on surface-level revisions, and emphasized technical constraints.

By linking learners' strategic engagement with GenAI to SRL processes, the study highlights how self-regulation shapes effective AI-mediated writing. These insights inform three pedagogical principles for optimizing learner–AI collaboration: aligning AI use with writing phases, fostering critical evaluation and revision of AI output, and addressing attributional orientations to promote adaptive regulation.

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Data availability statement

The data used in this study are available from the corresponding author upon reasonable request.

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